CSE 4733: Digital Image Processing

Chapter 7 — Wavelets and Multiresolution Processing

(Prepared from the 3rd Edition)

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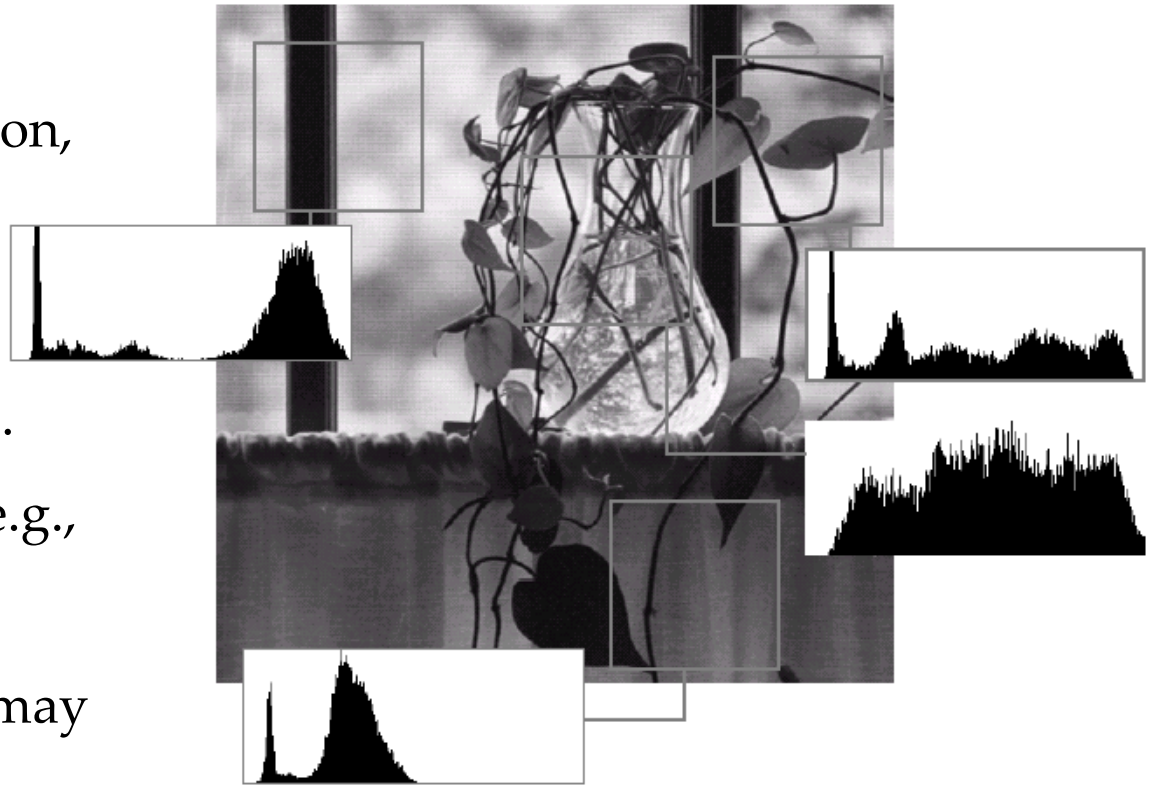
Introduction

Why is it important to have MRA

- ✓ Unlike Fourier transform, whose basis functions are sinusoids, wavelet transforms are based on small waves, called wavelets, of limited duration.
- ✓ Fourier transform provides only frequency information, but wavelet transform provides time-frequency information.
- ✓ Wavelets lead to a multiresolution analysis of signals.
- ✓ Multiresolution analysis: representation of a signal (e.g., an images) in more than one resolution/scale.
- ✓ Features that might go undetected at one resolution may be easy to spot in another.



Wavelets

What are they?

Wavelets are small waves of *varying frequency* and *limited duration*.

✓ **Trivia:** The word *wavelet* (from the French *ondelettes*) is first used by Grossman and Morlet in 1982 in the field of geophysics, when they develop the Continuous Wavelet Transform (CWT). [1]

While the Fourier functions (sines and cosines) are oscillating infinitely, the wavelets are functions with time localization. This means,

- ***Fourier Transforms*** — only *frequency* information.
- ***Wavelet Transforms*** — both *frequency* and *temporal* information.

e.g. think about music!

Wavelets are the foundation for ***Multiresolution theory***.

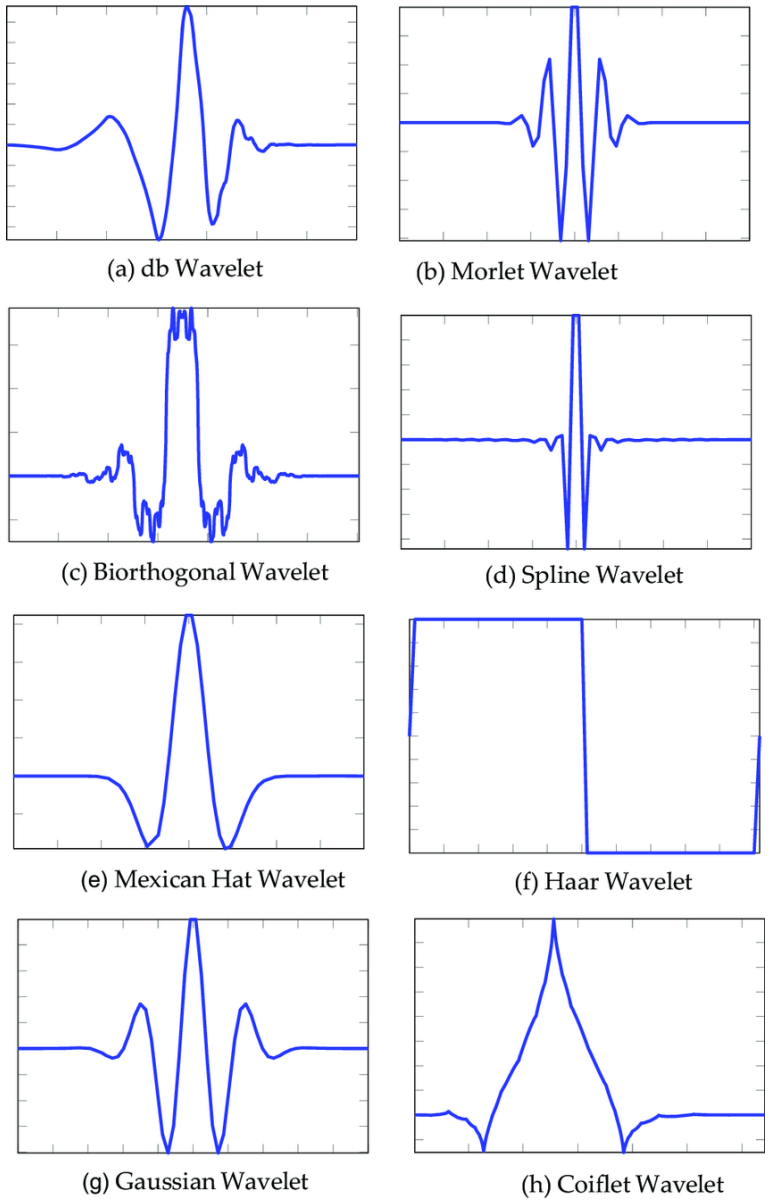


Fig: Wavelet examples.
Adapted from [3]

Multiresolution Theory

A brief synopsis

Multiresolution theory is concerned with the representation and analysis of signals (or images) at more than one resolution.

- ✓ **Utility:** To detect features at a particular resolution that might go undetected at another resolution.
- ✓ **Methods** and **techniques** under its umbrella —
  - Subband Coding (from digital signal processing)
  - Quadrature Mirror Filtering (from digital speech recognition)
  - Pyramidal Image Processing (from digital image processing)
  - and many many more…

In the domain of image processing, the initial impetus in multiresolution analysis research was in the late 1980s.

Image Pyramids

Understanding the concept

We can form an abstract hierarchy of representation of images at multiple resolutions — a pyramid with a collection of decreasing resolution images.

Conventionally, we consider P reduced resolution approximations of the original image.

In aggregate, the pixel count of a $P + 1$ level pyramid is,

$$N^2\left(1 + \frac{1}{(4)^1} + \frac{1}{(4)^2} + \cdots + \frac{1}{(4)^P}\right) \leq \frac{4}{3}N^2$$

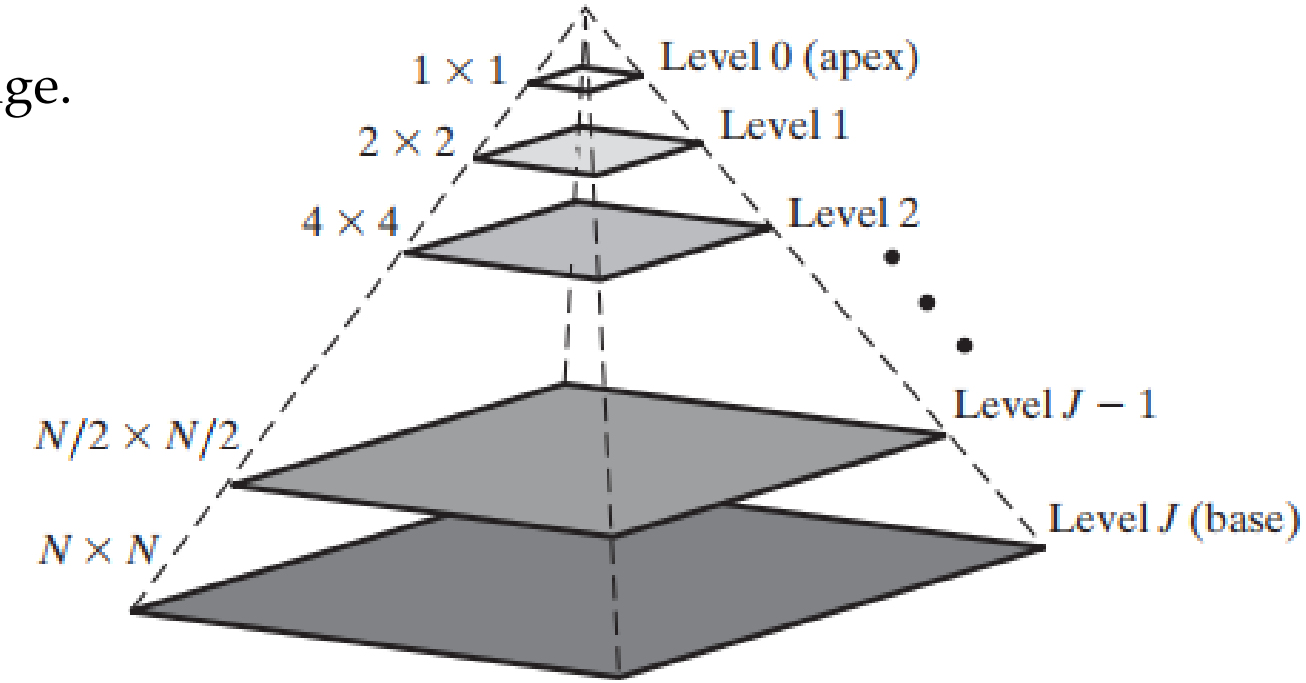


Fig: An image pyramid of an $N \times N$ image, where $N = 2^J$.

Image Pyramids

How to construct them?

Both *approximation* and *prediction residual* pyramids are computed in an iterative fashion.

The steps are delineated below —

- Compute a reduced-resolution approximation of the Level j input image. Do it by filtering and downsampling the filtered result by a factor of 2. Place the resulting approximation at level $j-1$ of the approximation pyramid.

$$f_{2\downarrow}(n) = f(2n)$$

- Create an estimate of the Level j input image from the approximation generated in step 1. Do it by upsampling and filtering.

- Compute the difference between the prediction image of step 2 and the input to step 1. Place it in level j of the prediction residual pyramid.

$$f_{2\uparrow}(n) = \begin{cases} f(n/2) & \text{if } n \text{ is even} \\ 0 & \text{otherwise} \end{cases}$$

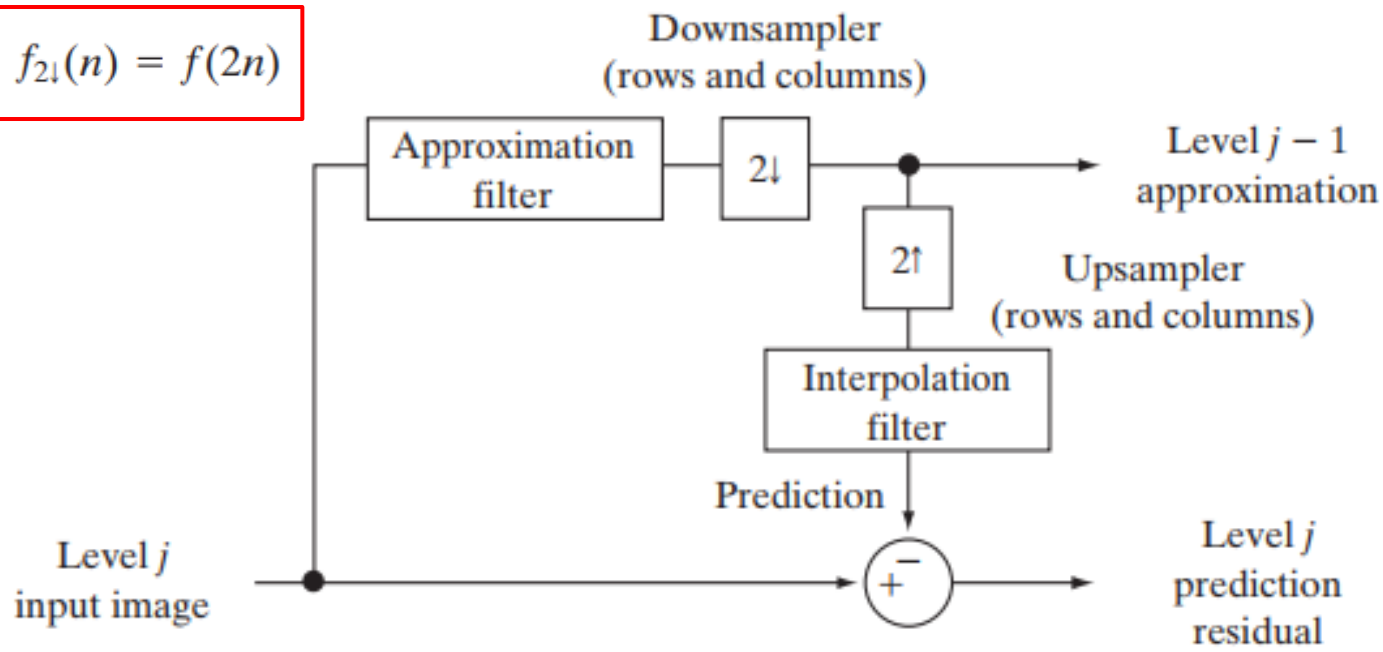


Fig: Process of creating approximation and prediction residual pyramids.

Image Pyramids

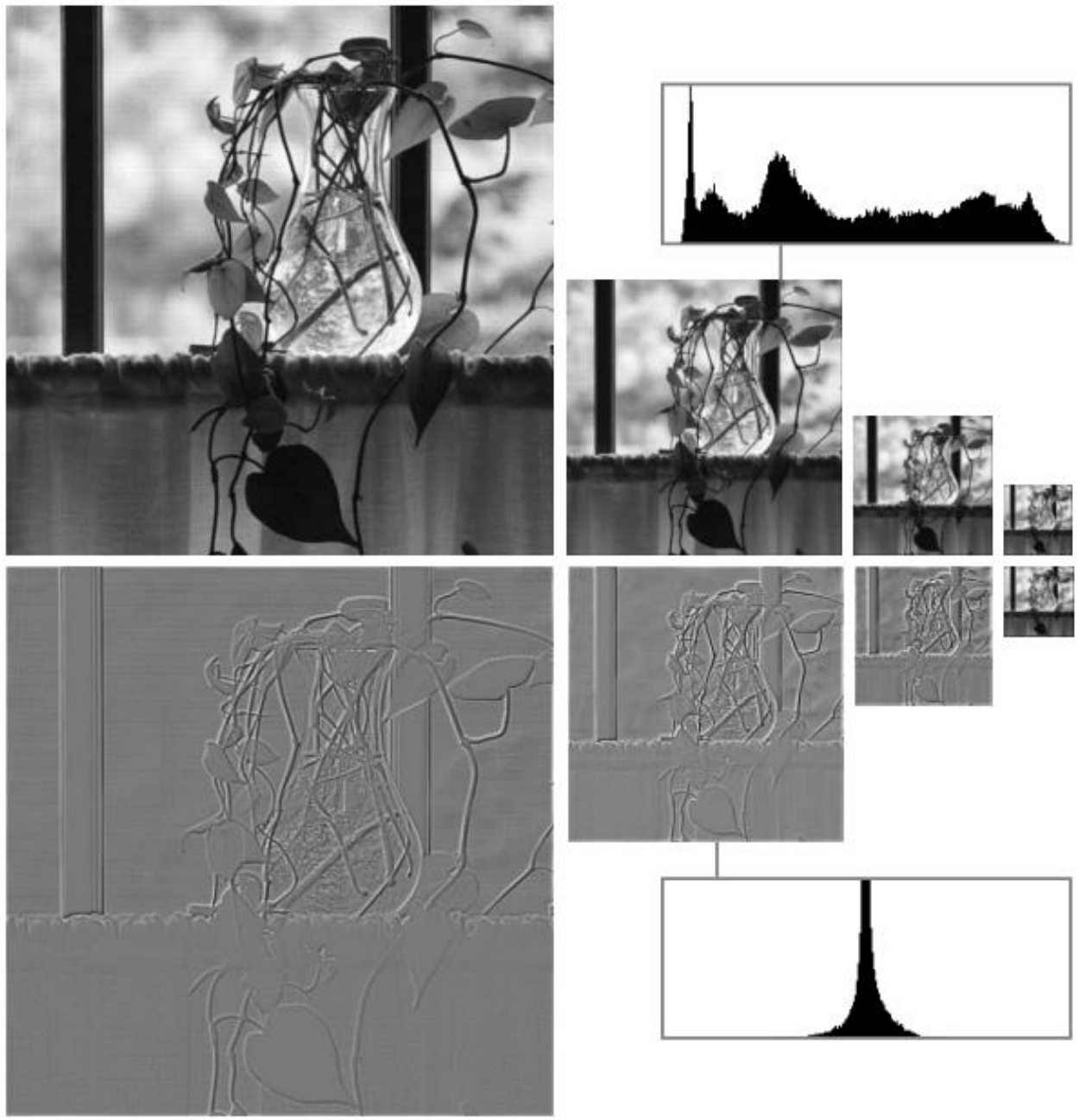
An example

Approximations

Fig: Two image pyramids and their histograms: an approximation pyramid; a prediction residual pyramid. The approximation pyramid is called a *Gaussian pyramid* because a Gaussian filter was used to construct it. The prediction residual pyramid is often called a *Laplacian Pyramid*.

Residuals

Takes *fewer* bits to store. Can leverage this to save memory!



Subband Coding

Understanding the concept

In subband coding, an image is decomposed into a set of bandlimited components, called ***subbands***. The decomposition is performed so that the subbands can be reassembled to reconstruct the original image ***without error***.

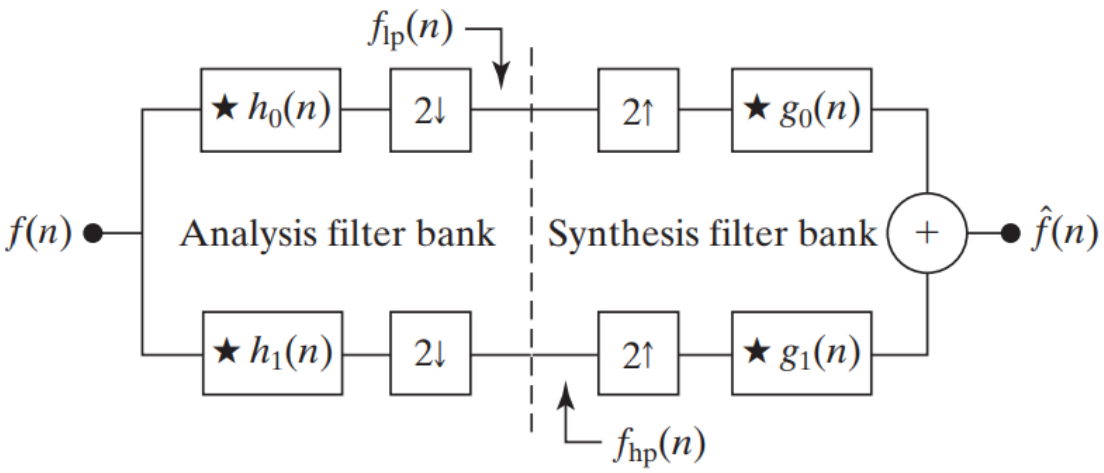


Fig: A two-band subband coding and decoding system.

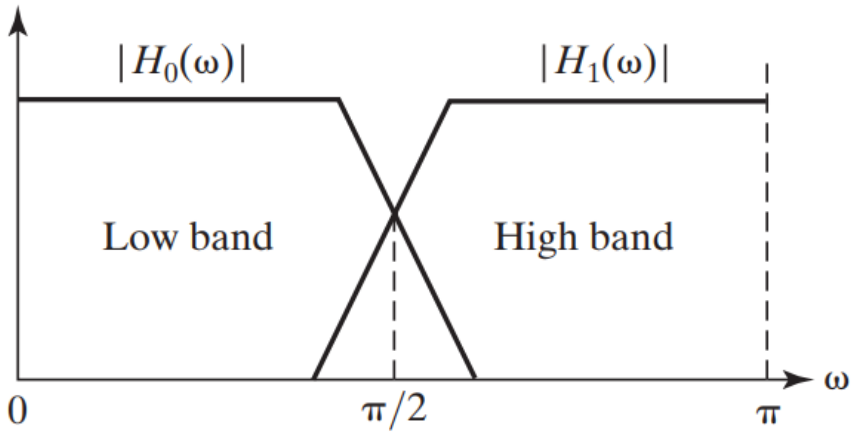


Fig: The spectrum splitting properties of the system.

Analysis filter bank — $h_0(n)$ and $h_1(n)$, for partitioning the input into low and high frequency subbands.

Synthesis filter bank — $g_0(n)$ and $g_1(n)$, for recombining the low and high frequency subbands to form $\hat{f}(n)$.

If $\hat{f}(n) = f(n)$, then the system is dubbed as *perfect reconstruction filters*.

Subband Coding

Conditions for perfect reconstruction

For *perfect reconstruction*, the impulse responses of the synthesis and analysis filters must be related in one of the following two ways:

$$g_0(n) = (-1)^n h_1(n) \qquad g_1(n) = (-1)^{n+1} h_0(n)$$

or

$$g_0(n) = (-1)^{n+1} h_1(n) \qquad g_1(n) = (-1)^n h_0(n)$$

For $h_0(n)$, $h_1(n)$, $g_0(n)$, and $g_1(n)$ are said to be ***cross-modulated*** because diagonally opposed filters.

These filters also abide by the ***biorthogonality*** condition,

$$\langle h_i(2n-k), g_j(k)\rangle = \delta(i-j)\delta(n), \quad i, j = \{0, 1\}$$

And the ***orthonormality*** conditions,

$$\langle g_i(n), g_j(n+2m)\rangle = \delta(i-j)\delta(m), \quad i, j = \{0, 1\}$$

If satisfied, take one as ***prototype*** and generate the rest!

Modulation

$$g_1(n) = \boxed{(-1)^n} g_0\boxed{(K_{\text{even}} - 1 - n)}$$

$$h_i(n) = g_i(K_{\text{even}} - 1 - n), \quad i = \{0, 1\}$$

Order reversal and **shifting**

Subband Coding

The 2D version ***(for images)***

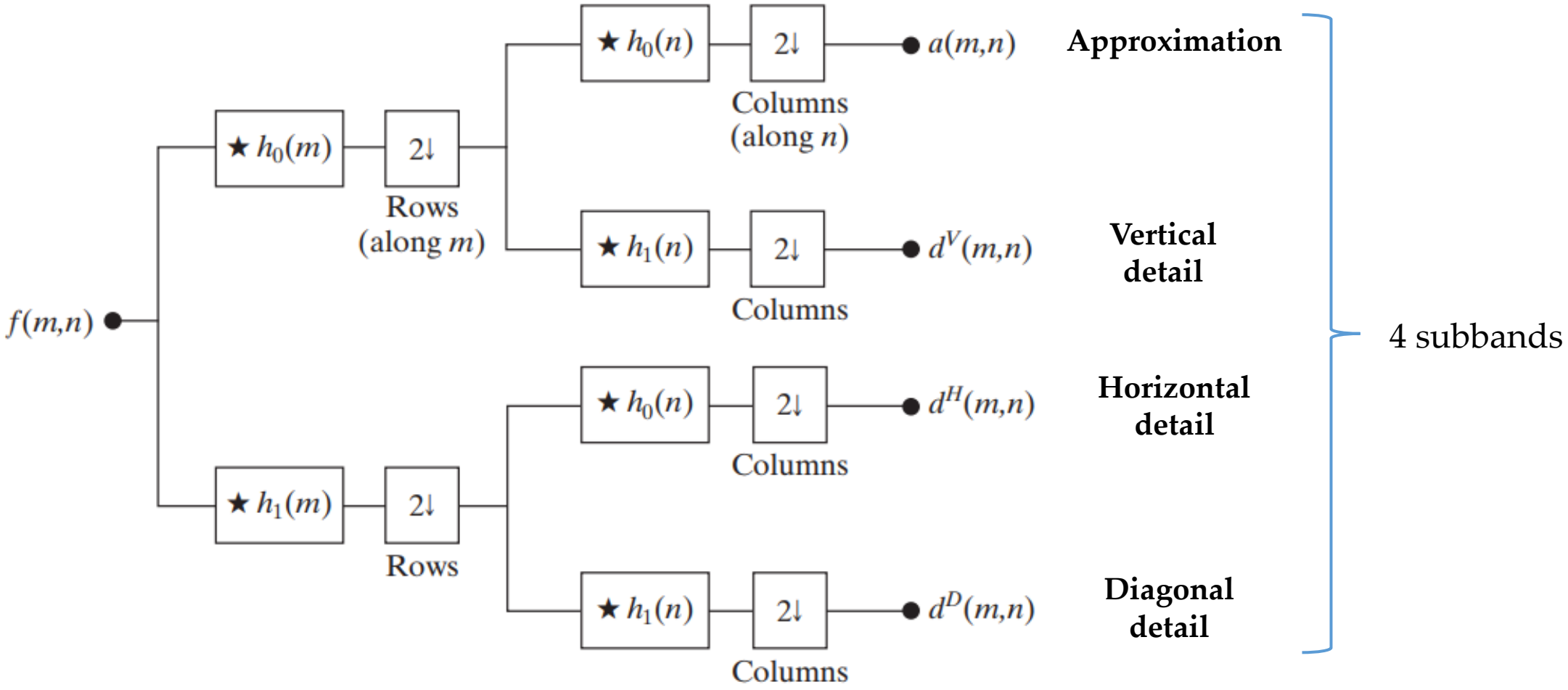


Fig: A two-dimensional, 4-band filter bank subband coding system.

Subband Coding

An example

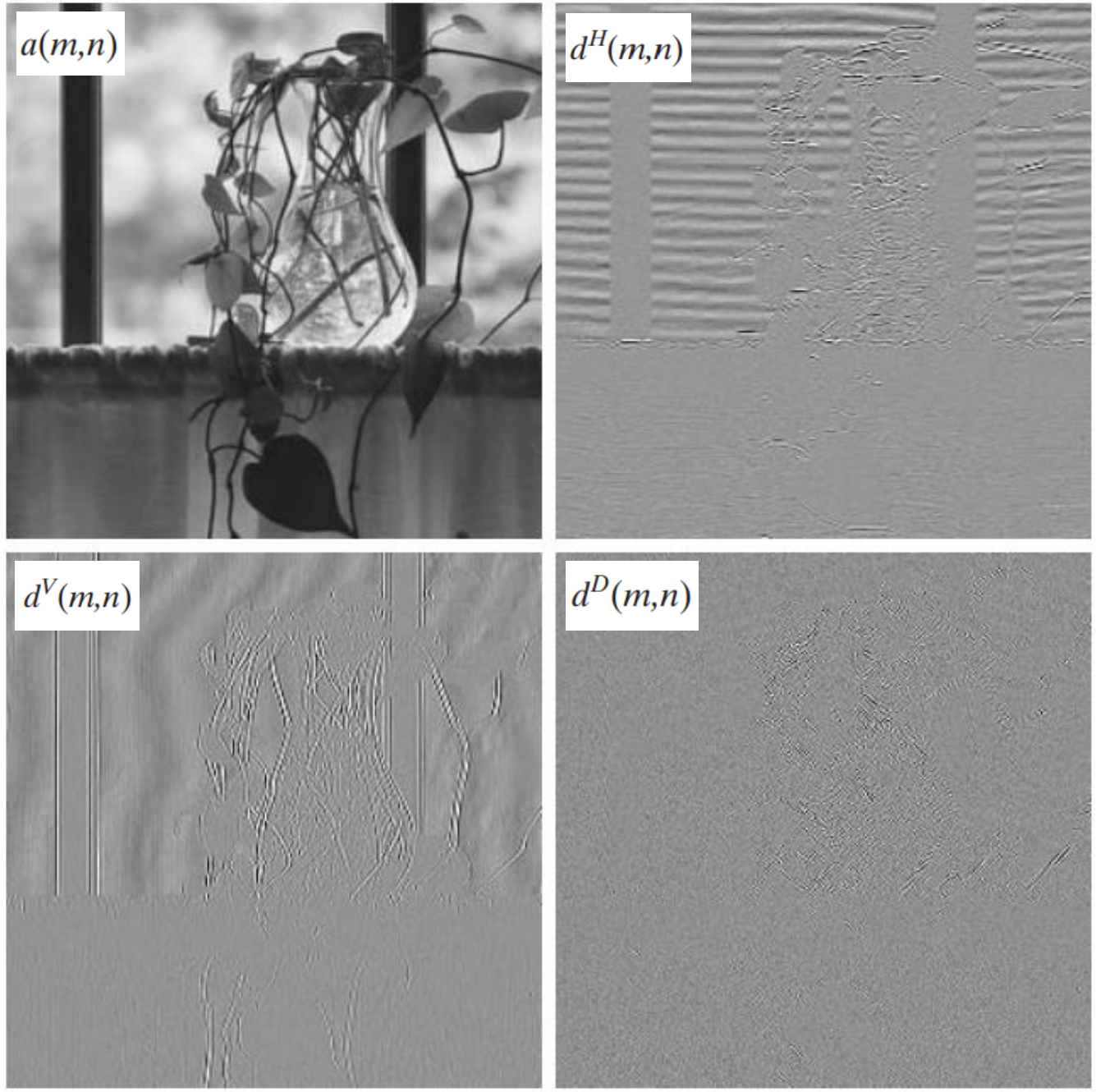


Fig: A four-band split of an image using the two-dimensional, four-band filter bank subband image coding system. The four subbands that result are the approximation, horizontal detail, vertical detail, and diagonal detail subbands

The Haar Transform

History

- **Haar proposed the Haar Transform in 1910, more than 70 years before the wavelet theory was born.**
- **Actually, Haar Transform employs the Haar wavelet filters but is expressed in a matrix form.**
- **Haar wavelet is the oldest and simplest wavelet basis.**
- **Haar wavelet is the only one wavelet basis, which holds the properties of orthogonal, (anti-)symmetric and compactly supported.**

Haar Transform

What is it?

The basis functions of the Haar transform are the ***oldest*** and ***simplest*** known orthonormal wavelets.

$$\mathbf{T} = \mathbf{H}\mathbf{F}\mathbf{H}^T$$

resulting $N \times N$ transform

$N \times N$ image matrix

$N \times N$ Haar transformation matrix

For the Haar transform, **H** contains the Haar basis functions $h_k(z)$, defined over the continuous, closed interval $z \in [0,1]$ for $k = 0, 1, 2, \ldots, N-1$, where $N = 2^n$.

To generate **H**, we take $k = 2^p + q - 1$, where $0 \le p \le n-1$, $q = 0$ or 1 for $p = 0$, and $1 \le q \le 2^p$ for $p \ne 0$.

$$h_0(z) = h_{00}(z) = \frac{1}{\sqrt{N}}, \quad z \in [0, 1]$$

$$h_k(z) = h_{pq}(z) = \frac{1}{\sqrt{N}} \begin{cases} 2^{p/2} & (q-1)/2^p \le z < (q-0.5)/2^p \\ -2^{p/2} & (q-0.5)/2^p \le z < q/2^p \\ 0 & \text{otherwise, } z \in [0, 1] \end{cases}$$

The ith row of an $N \times N$ Haar transformation matrix contains $h_i(z)$ for $z = \frac{0}{N}, \frac{1}{N}, \frac{2}{N}, \ldots, \frac{N-1}{N}$.

Haar Transform

Some example matrices

$$\mathbf{H}_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Smoothing

Gradient/ Difference

Normalizing factor

$$\mathbf{H}_4 = \frac{1}{\sqrt{4}}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ \sqrt{2} & -\sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{2} & -\sqrt{2} \end{bmatrix}$$

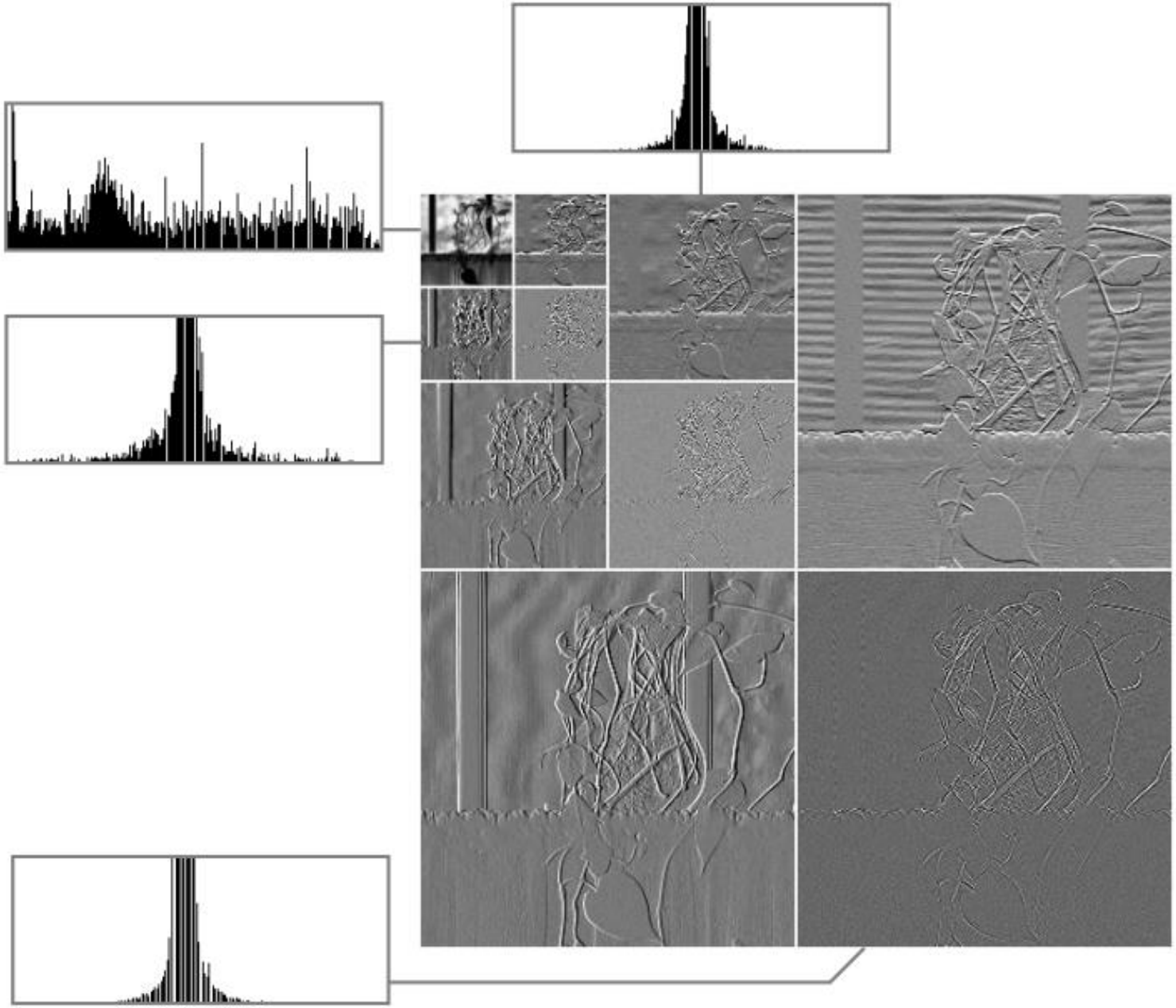
$$\mathbf{H}_8 = \frac{1}{\sqrt{8}}\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} \\ 2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 \end{bmatrix}$$

Haar Transform

Application in DWT (using H_2)

Coarse representation

Fine representation



Multiresolution Expansions

An overview

The goal of ***Multiresolution Analysis (MRA)*** is to create a series of approximations of a signal or image.

- The series approximations have a discerning ***factor of 2***.
- *Wavelets* are used to encode the difference in information between consecutive approximations of a signal or image.
  - Analysis of a signal over a large scale provides information about the gross features, *i.e.* the ***low frequencies***.
  - On the other hand, analysis with a small window (translated over the entire data set in steps) provides information about the ***high frequencies*** (*i.e.* the small scale variations or sharp discontinuities) and also localizes them [4].

The result will be a collection of time-frequency representations of the signal, all with different resolutions [5].

Series Expansions

The general idea

We can decompose a function f of variable x as a ***linear combination*** of expansion functions,

$$f(x) = \sum_k \alpha_k \varphi_k(x)$$

Where k is an integer index of a finite or infinite sum, the α_k are real-valued ***expansion coefficients***, and the $\varphi_k(x)$ are real-valued ***expansion functions***.

The expressible functions form a ***function space*** that is referred to as the ***closed span*** of the expansion set, denoted

$$V = \overline{\underset{k}{\mathrm{Span}}\{\varphi_k(x)\}}$$

The coefficients are the integral inner products of the dual $\hat{\varphi}_k(x)$ and function $f(x)$.

$$\alpha_k = \langle \tilde{\varphi}_k(x), f(x) \rangle = \int \boxed{\tilde{\varphi}_k^*(x)} f(x)\, dx$$

Complex conjugate of the dual

Series Expansions

3 cases to consider for calculating the coefficients

Case 1: If the expansion functions form an ***orthonormal basis*** for meaning that

$$\langle \varphi_j(x), \varphi_k(x) \rangle = \delta_{jk} = \begin{cases} 0 & j \neq k \\ 1 & j = k \end{cases}$$

$$\alpha_k = \langle \varphi_k(x), f(x) \rangle$$

Indicates that the basis and its dual are equivalent

$$\varphi_k(x) = \tilde{\varphi}_k(x)$$

Case 2: If the expansion functions are ***not orthonormal***, but are an orthogonal basis for V, then

$$\langle \varphi_j(x), \varphi_k(x) \rangle = 0 \quad j \neq k \qquad \langle \varphi_j(x), \tilde{\varphi}_k(x) \rangle = \delta_{jk} = \begin{cases} 0 & j \neq k \\ 1 & j = k \end{cases}$$

Case 3: If the expansion set is not a basis for V, but supports the series expansion, then there can be many α_k values. They form a ***frame*** based on $A\|f(x)\|^2 \leq \sum_k |\langle \varphi_k(x), f(x) \rangle|^2 \leq B\|f(x)\|^2$ which becomes ***tight*** [6] when $f(x) = \frac{1}{A} \sum_k \langle \varphi_k(x), f(x) \rangle \varphi_k(x)$

Scaling Functions

What are they?

These are expansion functions with 2 parameters — binary scaling j and integer translation k.

$$\boxed{\varphi_{j,k}(x) = 2^{j/2}\varphi(2^j x - k)}$$

$j, k \in \mathbb{Z}$

For an appropriate j_0, spans the whole of $L^2(\boldsymbol{R})$

$$V_{j_0} = \overline{\operatorname*{Span}_k\{\varphi_{j_0,k}(x)\}}$$

Haar Scaling Function Properties: real, square-integrable *i.e.* $\varphi_{j,k}(x) \in L^2(\boldsymbol{R})$

$$\varphi(x) = \begin{cases} 1 & 0 \le x < 1 \\ 0 & \text{otherwise} \end{cases}$$

That is, V_{j_0} is the span of $\varphi_{j_0,k}(x)$ over k. If $f(x) \in V_{j_0}$, then, $f(x) = \sum_k \alpha_k \varphi_{j_0,k}(x)$

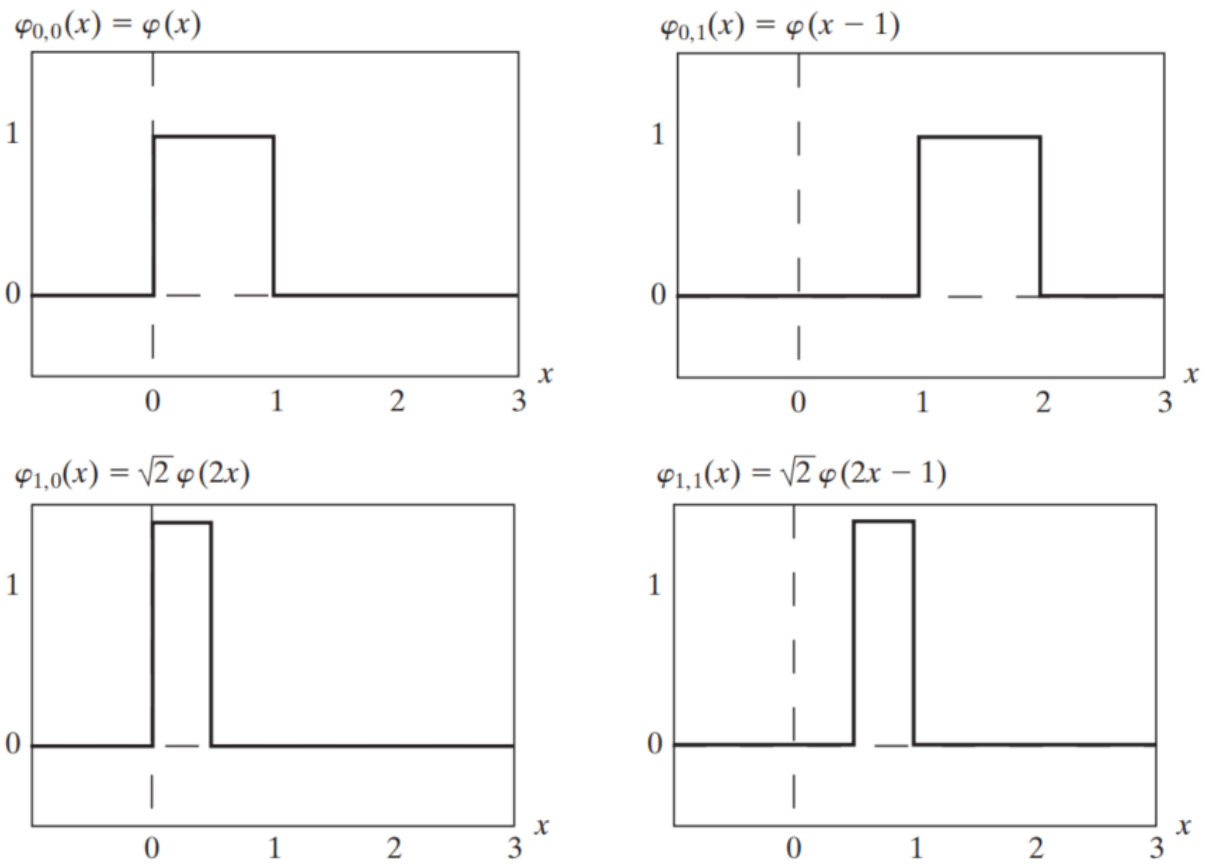
If j increases, it can represent finer details and smaller variations, since the $\varphi_{j,k}(x)$ functions become narrower and are separated by small changes in x.

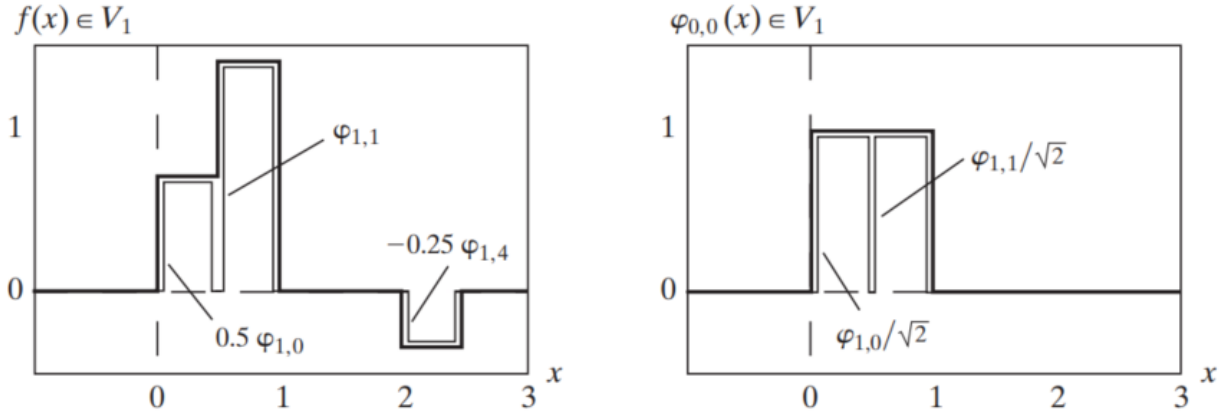
Scaling Functions

Some examples

$$\varphi_{0,k}(x) = \frac{1}{\sqrt{2}}\varphi_{1,2k}(x) + \frac{1}{\sqrt{2}}\varphi_{1,2k+1}(x)$$

$$V_0 \subset V_1$$





$$f(x) = 0.5\varphi_{1,0}(x) + \varphi_{1,1}(x) - 0.25\varphi_{1,4}(x)$$

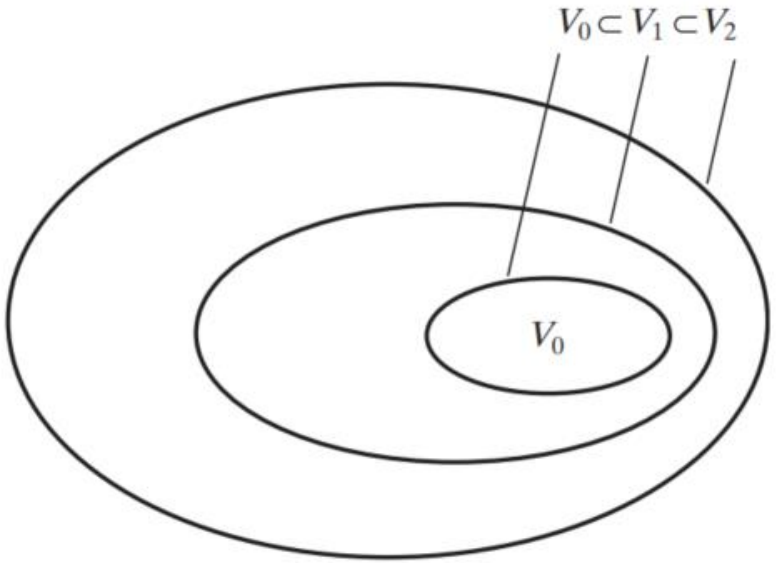
Scaling Functions

The MRA requirements they obey

MRA Requirement 1: The scaling function is orthogonal to its integer translates.

MRA Requirement 2: The subspaces spanned by the scaling function at low scales are nested within those spanned at higher scales.

$$V_{-\infty} \subset \cdots \subset V_{-1} \subset V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_{\infty}$$



MRA Requirement 3: The only function that is common to all V_j is $f(x) = 0$.

MRA Requirement 4: Any function can be represented with arbitrary precision.

$$V_{\infty} = \{L^2(\boldsymbol{R})\}$$

Scaling Functions

When the MRA requirements are satisfied

The expansion functions of subspace V_j can be expressed as a weighted sum of the expansion functions of subspace V_{j+1}.

$$\varphi_{j,k}(x) = \sum_n \alpha_n \varphi_{j+1,n}(x)$$

$$\varphi_{j,k}(x) = \sum_n h_\varphi(n) 2^{(j+1)/2} \varphi(2^{j+1}x - n)$$

Scaling functions coefficients where h_φ is the **scaling vector**

$$\varphi(x) = \sum_n h_\varphi(n) \sqrt{2} \varphi(2x - n)$$

Setting j and k to 0 and since $\varphi(x) = \varphi_{0,0}(x)$

The ***Refinement equation*** or ***MRA equation*** or ***Dilation equation***

■ The scaling function coefficients for the Haar function of Eq. (7.2-14) are $h_\varphi(0) = h_\varphi(1) = 1/\sqrt{2}$, the first row of matrix $\mathbf{H}_2$ in Eq. (7.1-18). Thus, Eq. (7.2-18) yields

$$\varphi(x) = \frac{1}{\sqrt{2}}\left[\sqrt{2}\varphi(2x)\right] + \frac{1}{\sqrt{2}}\left[\sqrt{2}\varphi(2x - 1)\right]$$

Wavelet Functions

What are they?

It spans the difference between any two adjacent scaling subspaces V_j and V_{j+1}. It is denoted using $\psi(x)$ and has the same parameters as scaling functions j and k. We define a set of wavelets $\{\psi_{j,k}(x)\}$,

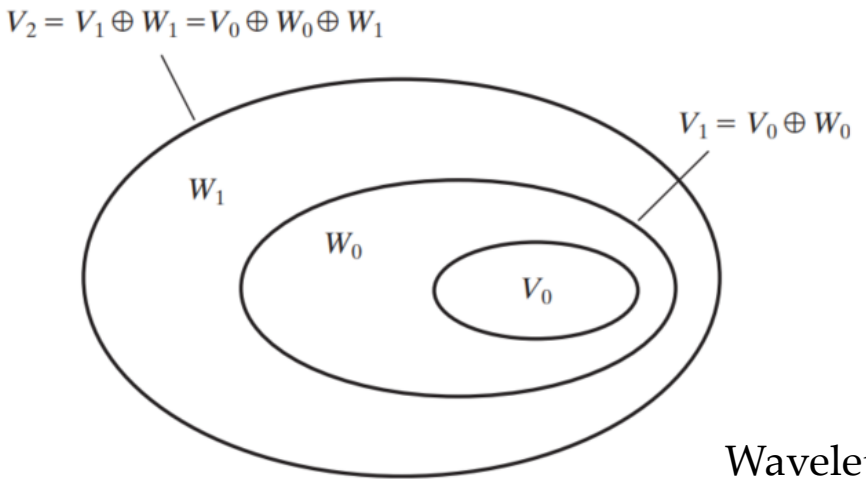
$j, k \in \mathbb{Z}$

$$\psi_{j,k}(x) = 2^{j/2}\psi(2^j x - k)$$

$$\psi(x) = \sum_n h_\psi(n)\sqrt{2}\varphi(2x - n)$$

$$W_j = \overline{\operatorname{Span}_k\{\psi_{j,k}(x)\}}$$

$$V_{j+1} = V_j \oplus W_j$$



$$\psi(x) = \begin{cases} 1 & 0 \le x < 0.5 \\ -1 & 0.5 \le x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

If $f(x) \in W_j$, then, $f(x) = \sum_k \alpha_k \psi_{j,k}(x)$

$$L^2(\mathbf{R}) = V_0 \oplus W_0 \oplus W_1 \oplus \ldots$$

$$L^2(\mathbf{R}) = V_1 \oplus W_1 \oplus W_2 \oplus \ldots$$

$$L^2(\mathbf{R}) = \cdots \oplus W_{-2} \oplus W_{-1} \oplus W_0 \oplus W_1 \oplus W_2 \oplus \ldots$$

$$L^2(\mathbf{R}) = V_{j_0} \oplus W_{j_0} \oplus W_{j_0+1} \oplus \ldots$$

Wavelet functions coefficients where h_ψ is the wavelet vector

$$h_\psi(n) = (-1)^n h_\varphi(1 - n)$$

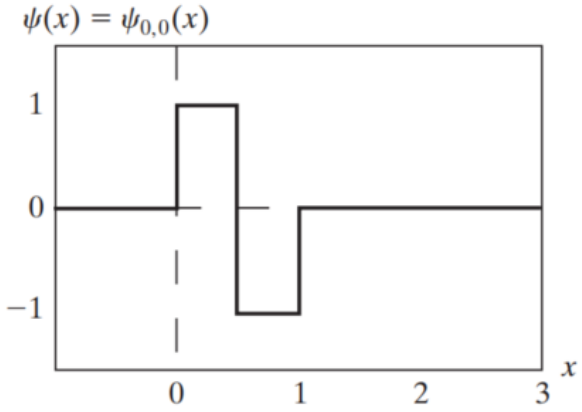
Wavelet Functions

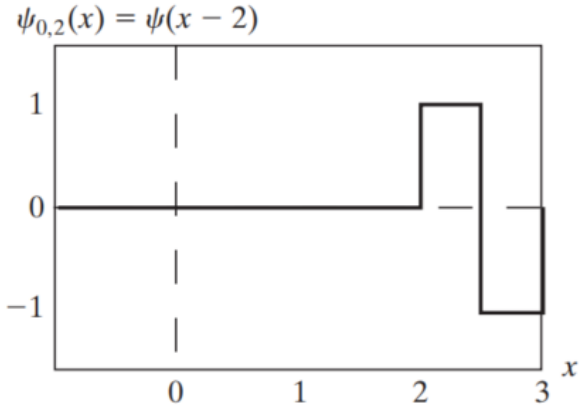
Some examples

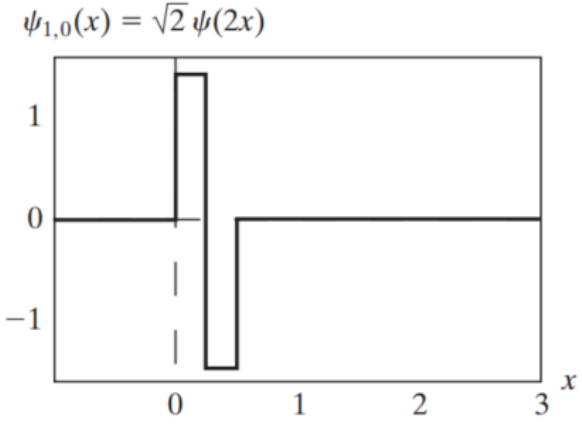
$$f(x) = f_a(x) + f_d(x)$$

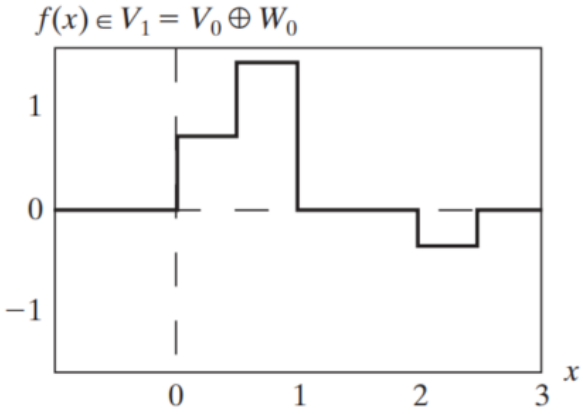
$f_a(x)$ is an approximation of $f(x)$ using V_0 scaling functions

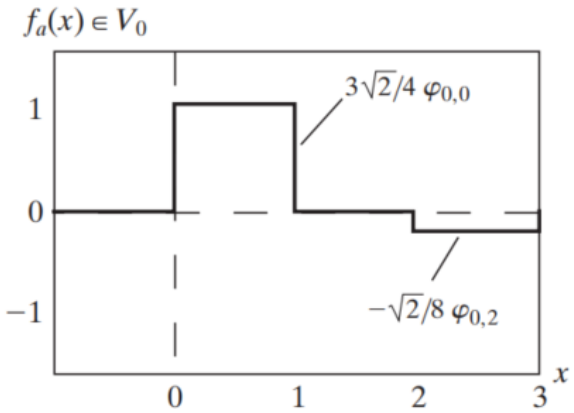
$f_d(x)$ is the difference $f(x) - f_a(x)$ as a sum of W_0 wavelets

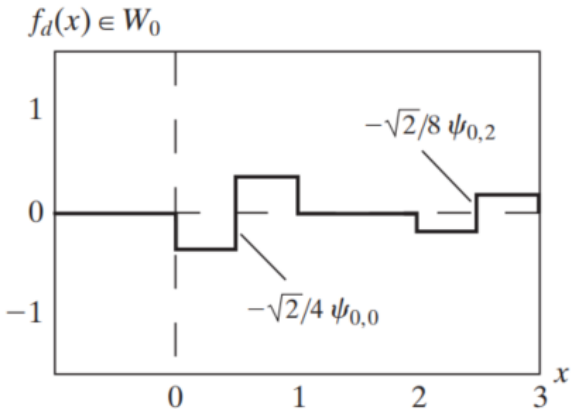












$$f_a(x) = \frac{3\sqrt{2}}{4}\varphi_{0,0}(x) - \frac{\sqrt{2}}{8}\varphi_{0,2}(x)$$

$$f_d(x) = \frac{-\sqrt{2}}{4}\psi_{0,0}(x) - \frac{\sqrt{2}}{8}\psi_{0,2}(x)$$

Wavelet Transforms in One Dimension

An introduction

Wavelet Transforms

Wavelet Series Expansion

Discrete Wavelet Transform

Continuous Wavelet Transform

Wavelet Series Expansions

What is it?

The *wavelet series expansion* of function $f(x) \in L^2(\mathbf{R})$ relative to wavelet $\psi(x)$ and scaling function $\varphi(x)$ can be represented by a scaling function expansion in subspace V_{j_0} and some wavelet expansions in the subspaces $W_{j_0}, W_{j_0+1}, \ldots$

$$f(x) = \boxed{\sum_k c_{j_0}(k)\varphi_{j_0,k}(x)} + \boxed{\sum_{j=j_0}^{\infty}\sum_k d_j(k)\psi_{j,k}(x)}$$

Rough approximation of $f(x)$ in at scale j_0

Fine details at each higher scale $j \geq j_0$

Wavelet Series Expansions

The expansion coefficients

The expansion coefficients are the ***inner products*** of the function being expanded and the expansion function being used.

Approximation coefficient

$$\boxed{c_{j_0}(k)} = \langle f(x), \varphi_{j_0,k}(x) \rangle = \int f(x)\varphi_{j_0,k}(x)\,dx$$

Detail coefficient

$$\boxed{d_j(k)} = \langle f(x), \psi_{j,k}(x) \rangle = \int f(x)\psi_{j,k}(x)\,dx$$

<u>Condition:</u> The expansion functions should form an ***orthonormal basis***. (In vector terms, they are ***linearly independent***, ***perpendicular***, of ***unit size***, and ***span the entire vector space***.)

<u>Caveat:</u> If they belong to a ***biorthogonal basis***, we need to use their respective ***dual functions***.

Wavelet Series Expansions

A simple example

Using ***Haar wavelets*** and considering $j_0 = 0$, let's calculate the WSE of —

$$y = \begin{cases} x^2 & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$c_0(0) = \int_0^1 x^2 \varphi_{0,0}(x)\, dx = \int_0^1 x^2\, dx = \left.\frac{x^3}{3}\right|_0^1 = \frac{1}{3}$$

$$d_0(0) = \int_0^1 x^2 \psi_{0,0}(x)\, dx = \int_0^{0.5} x^2\, dx - \int_{0.5}^1 x^2\, dx = -\frac{1}{4}$$

$$d_1(0) = \int_0^1 x^2 \psi_{1,0}(x)\, dx = \int_0^{0.25} x^2\sqrt{2}\, dx - \int_{0.25}^{0.5} x^2\sqrt{2}\, dx = -\frac{\sqrt{2}}{32}$$

$$d_1(1) = \int_0^1 x^2 \psi_{1,1}(x)\, dx = \int_{0.5}^{0.75} x^2\sqrt{2}\, dx - \int_{0.75}^1 x^2\sqrt{2}\, dx = -\frac{3\sqrt{2}}{32}$$

Recall that,

$$\varphi(x) = \begin{cases} 1 & 0 \le x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\psi(x) = \begin{cases} 1 & 0 \le x < 0.5 \\ -1 & 0.5 \le x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$\psi_{j,k}(x) = 2^{j/2}\psi(2^j x - k)$$

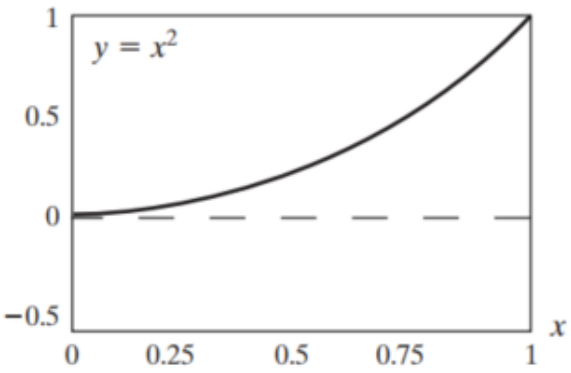
Wavelet Series Expansions

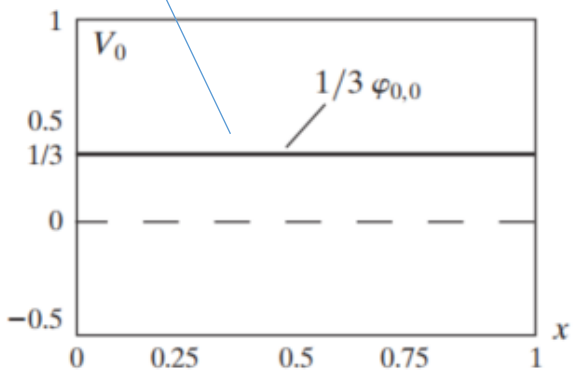
A simple example

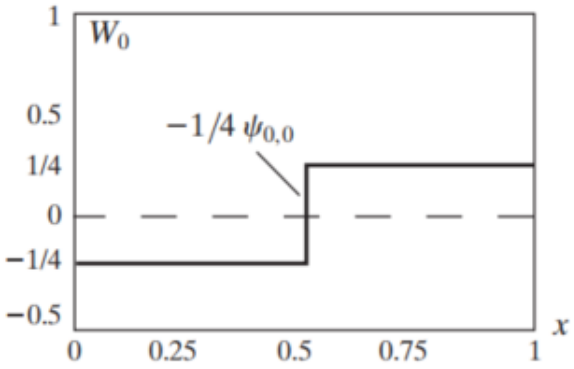
Substituting these values into Eq. (7.3-1), we get the wavelet series expansion

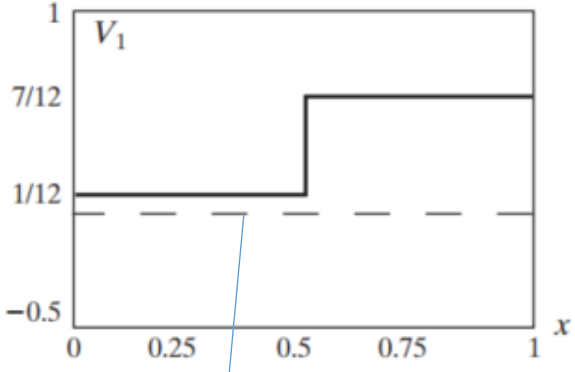
$$y = \underbrace{\underbrace{\underbrace{\frac{1}{3}\varphi_{0,0}(x)}_{V_0} + \underbrace{\left[-\frac{1}{4}\psi_{0,0}(x)\right]}_{W_0}}_{V_1 = V_0 \oplus W_0} + \underbrace{\left[-\frac{\sqrt{2}}{32}\psi_{1,0}(x) - \frac{3\sqrt{2}}{32}\psi_{1,1}(x)\right]}_{W_1}}_{V_2 = V_1 \oplus W_1 = V_0 \oplus W_0 \oplus W_1} + \cdots$$

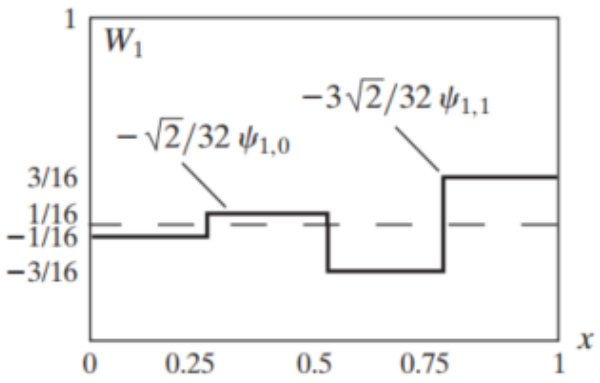
Average of the function

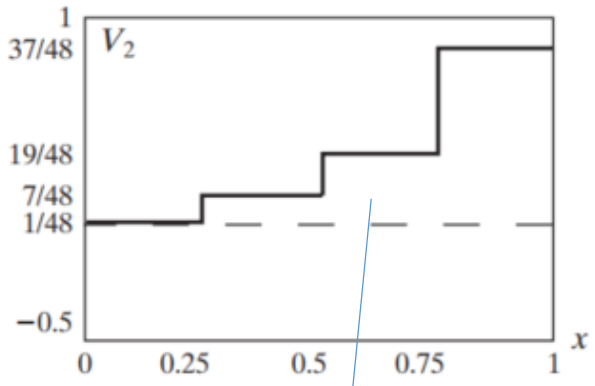












Bad V_1 level approx.

Somewhat good V_2 level approx.

As $j \to \infty$, the approximation get ***more and more precise***!